Exploratory Data Analysis (EDA) Report

Project Title

Sales Data Analysis and Business Intelligence Dashboard

Objective

The objective of this project is to analyze sales data using Python, SQL, and data visualization techniques to identify business trends, customer behavior, product performance, and profitability.

Dataset Overview

The dataset contains sales transaction records including:

- Product
- Customer Name
- Sales
- Profit

A total of **90 orders** were analyzed.

Data Exploration

The dataset was loaded into Python using Pandas and explored using the following functions:

- df.head()
- df.info()
- df.describe()
- df.isnull().sum()

Findings

- No major missing values were observed.
- Numerical columns include Sales and Profit.
- Categorical columns include Product and Customer Name.

Descriptive Statistics

Total Orders

Metric	Value
Total Orders	90

Average Sales

Metric	**Value**
Average Sales	27,357.95

Highest Revenue Product

Product
Webcam

Visual Analysis

1. Revenue Share by Product

A pie chart was created to analyze the revenue contribution of each product.

Observation

- Webcam contributed the highest revenue share.
- SSD and Headphones also contributed significantly.
- Mouse generated comparatively lower revenue.

---

2. Profit by Product

A horizontal bar chart was used to compare profits generated by products.

Observation

Product	**Profit**
Webcam	88,082.39
SSD	85,417.82
Headphones	66,004.21
Laptop	65,343.05
Mouse	58,084.99

Insight

Webcam generated the highest profit among all products.

---

3. Top Customers by Sales

A bar chart was created to identify the highest contributing customers.

Observation

Customer	**Sales**
Rohit	303,606.92
Priya	299,084.36
Kiran	281,994.62
Pooja	276,533.42
Meera	262,043.66

Insight

Rohit was the highest revenue-generating customer.

---

4. Revenue by Product

A column chart was used to compare product revenue.

Observation

Product	**Revenue**
Webcam	517,070.30
Headphones	400,160.82
SSD	382,114.42
Mouse	327,650.11
Laptop	306,992.35

Insight

Webcam was the best-selling and highest revenue-generating product.

---

5. Revenue vs Profit Analysis

A scatter plot was created to study the relationship between revenue and profit.

Insight

Products with higher sales generally generated higher profits. Webcam and SSD showed strong business performance in both revenue and profitability.

SQL Analysis

The following business questions were answered using SQL:

1. Top 5 Products by Revenue

2. Total Number of Orders
3. Product-wise Profit Analysis
4. Customer-wise Sales Analysis
5. Average Sales Value

SQL queries and outputs are provided in the SQL_Queries.md file.

Dashboard Summary

A static dashboard was created using Excel containing:

- KPI Cards
    - Total Orders
    - Average Sales
    - Highest Revenue Product
- Revenue Share by Product
- Profit by Product
- Top Customers by Sales
- Revenue by Product
- Revenue vs Profit

The dashboard provides a quick overview of business performance and key metrics.

Key Business Insights

1. Webcam is the highest revenue-generating product.
2. Webcam also generates the highest profit.
3. Rohit is the top customer based on total sales.
4. Average sales per order is 27,357.95.
5. Higher revenue products generally produce higher profits.
6. Product performance is concentrated among Webcam, SSD, and Headphones.

Conclusion

The analysis successfully identified top-performing products, profitable product categories, and valuable customers. The dashboard and SQL analysis provide actionable business insights that can support future sales and marketing decisions.